ONE-WAVE FRAMEWORK

Book 1 — Micro

Chapter 10: Time at Micro Scale — Counted Updates and Damping

Version: 4.0 Date: July 22, 2026 Class: B — Applied Layer Spine: Gray / 2D / 3D / Mathematics / Predictions / Yellow Audit / Future Work / Closing Thoughts

Dependencies: A-109 Inertial Memory, A-111 Recursion, C-309 Friction Limit, C-313 Lorentz Invariance Conflict, F-608 Attenuation Status: GREEN (time ontology and relativistic interpretation) / YELLOW (counted-step and damping mathematics)

Gray — Standard Physics Reference

In established physics, time is a coordinate in spacetime. Special and general relativity predict time dilation, and those predictions are experimentally confirmed. Quantum mechanics usually treats time as a parameter rather than an observable operator. The thermodynamic arrow of time is associated with entropy increase.

One-Wave does not yet derive the measured relativistic formulas from its lattice update rule. Those formulas remain comparison targets, not accomplishments of this chapter.

2D One-Wave Interpretation

The lattice evolves through ordered updates:

$$\psi^{n-1} \to \psi^{n} \to \psi^{n+1}.$$

The index `n` counts updates. If one update has duration `Delta t`, the elapsed lattice time after `N` updates is

$$t = N\Delta t.$$

Time count and state-change magnitude are not the same variable. Damping and coupling determine how much the state changes during a tick; they do not determine how many ticks have elapsed.

3D One-Wave Interpretation

The canonical update rule is

$$\psi_i^{n+1} = \psi_i^{n} + (1-\gamma)(\psi_i^{n} - \psi_i^{n-1}) + \beta_i(\langle \psi_j^{n} \rangle - \psi_i^{n}).$$

`gamma` suppresses inertial carry-forward. `beta` controls neighbor coupling. Neither coefficient alone is “time.”

The continuum damping rate is

$$\mu = \frac{\gamma}{\Delta t}.$$

The damping relaxation time is

$$\tau_d = \frac{1}{\mu} = \frac{\Delta t}{\gamma}.$$

This is distinct from elapsed time:

$$\boxed{t = N\Delta t \neq \frac{N}{\gamma}}.$$

The earlier `t=N/gamma` expression conflated elapsed update count with damping relaxation.

Arrow of Time

The notation `n -> n+1` establishes update order, but order notation alone does not prove physical irreversibility. The canonical continuum equation contains

$$\mu\psi_t,$$

which damps recoverable motion and selects a preferred time direction when `mu>0`. In the ideal `mu=0` limit, the wave equation can be time-reversal symmetric. Therefore the One-Wave arrow-of-time hypothesis depends on real damping or information loss, not merely on numbering the updates forward.

This creates the explicit C-313 tension: a damped preferred-frame equation is not exactly Lorentz invariant.

Motion and Gravitational Clock Rates

The current One-Wave hypothesis is that propagation, coupling, and internal change compete for finite update capacity. A moving or strongly coupled mode may therefore complete less internal change per external lattice tick.

That is still a Green mechanism sketch. The repository has not derived

$$t' = t\sqrt{1 - v^2/c^2}$$

or

$$t' = t\sqrt{1 - 2GM/(rc^2)}$$

from the canonical update rule. These remain required comparison targets. No ad hoc `gamma_eff` formula is retained as though it were a derivation.

Minimum Tick

If the lattice has a physical spatial step `Delta x` and maximum propagation speed `c`, a candidate minimum tick is

$$\Delta t_{\min} = \frac{\Delta x}{c}.$$

Identifying this with Planck time requires an independently derived `Delta x`; it is not established here.

Mathematics

Elapsed time

$$t_N = N\Delta t.$$

Damping rate and relaxation

$$\mu = \frac{\gamma}{\Delta t}, \qquad \tau_d = \frac{1}{\mu} = \frac{\Delta t}{\gamma}.$$

Finite continuum scaling

For a fixed physical damping rate as the timestep is refined:

$$\gamma = \mu\Delta t + O(\Delta t^2).$$

Damped propagation

$$\psi_{tt} + \mu\psi_t - c_{\text{eff}}^2\nabla^2\psi = 0.$$

For a Fourier mode:

$$\omega = -\frac{i\mu}{2} \pm \sqrt{c_{\text{eff}}^2 k^2 - \frac{\mu^2}{4}}.$$

This separates propagating, critical, and overdamped regimes. See C-313.

Attenuation comparison

F-608 uses the provisional spatial law

$$A(x) = A_0 e^{-\eta x}.$$

Its attenuation coefficient `eta` has not yet been derived from `gamma`, `beta`, and frequency. Spatial attenuation and temporal damping must not be silently treated as identical.

Predictions and Tests

1. If time is counted updates, any physical minimum tick requires a measurable or derivable lattice step.
2. If damping supplies the arrow of time, reducing `mu` should increase reversibility and memory recovery in simulations.
3. The weak-damping ratio `epsilon=mu/(c_eff k)` should control when propagation appears nearly undamped.
4. Relativistic time dilation must be derived from the same canonical variables without inserting the known square-root formulas by hand.
5. A successful model must distinguish elapsed tick count, damping relaxation, propagation phase, and internal clock change.

Yellow Audit

Completed:

- obsolete physics I-05 citation removed; real dependencies identified,
- legacy Friction Limit citation repointed to C-309,
- elapsed time separated from damping time,
- finite continuum scaling stated,
- arrow-of-time claim narrowed to damping/information loss.

Still open:

- derivation of special-relativistic time dilation,
- derivation of gravitational time dilation,
- physical calibration of `Delta t`, `Delta x`, `mu`, and `beta`,
- derivation linking temporal damping to spatial attenuation,
- experimental test of a preferred lattice frame.

Closing Thoughts

One-Wave's useful claim is not that `gamma` is time. It is that ordered recursive updates provide a count, while resistance and coupling determine how much change can occur during each count.

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time count = number of updates
damping time = loss/recovery timescale
clock rate = internal change per update
```

Those quantities may interact, but they are not interchangeable.

END OF BOOK 1 CHAPTER 10